

## EDUCATION

---

### PhD in Probabilistic Modeling for Healthcare Management and Operations Oct 2024 — Present

- Recipient of WGSSS-ESRC Studentship Award
- Project focused on enhancing discharge planning process using probabilistic modelling

### MSc in Financial Mathematics Oct 2017 — Aug 2021

- Dissertation: Effects of Exchange Rate Volatility and Firm-Specific Features on the Rates of Returns of the Manufacturing Firms Listed in Stock Market: A CAPM Approach
  - Statistical&Machine learning techniques used: Markov Switching GARCH Models, ARIMA, Panel Data Econometrics, Principal Component Analysis

### BSc in Business Administration Oct 2011 — Aug 2015

## TEACHING

---

### Graduate Tutor in Business Data Analytics Oct 2025 — Present

- Designed lab work materials and lesson plans for MSc-level courses on Business Data Analytics, utilizing Python and R programming.
- Conducted weekly lab sessions, guiding and assisting students through practical applications of data analysis techniques.

### Training sessions for a research group members

- Causal Inference via Machine Learning Feb 2026
- Introduction to Dynamic Programming and Reinforcement Learning May 2025
- Introduction to Python and Quarto on Visual Studio Code Mar 2025

## WORK EXPERIENCES

---

### Reporting and Data Analytics Executive at Commercial Bank Jun 2022 — Sep 2024

- Applied time-series forecasting and machine learning techniques, including ARIMA, Bayesian Time Series, Prophet, ANN, LSTM, Random Forest, LightGBM and XGBoost, to historical data for making long-horizon forecasts of daily customer call volume. The winning model helped to allocate resources effectively, reducing customer wait times by 6%.
- Implemented machine learning models (e.g., XGBoost, LightGBM, CatBoost) to predict customer behavior with an over recall rate of 70% and an accuracy rate of 85%, enabling fewer customer complaints and increase sales by 14%.
- Analyzing large amounts of data to identify trends and find patterns, signals and hidden stories within customer calls data.
- Applied machine learning techniques (Z-score, IsolationForest) to detect anomalies.
- Applied unsupervised machine learning techniques (DBSCAN, Gaussian mixture, K-means) to cluster customers.
- Hyperparameter tuning for machine learning models using Hyperopt, Optuna and KerasTuner.

### Research Associate at a Think Tank Jan 2022 — Apr 2024

- Determined areas of research to increase knowledge in the particular field.

- Utilizing inferential statistics such as hypothesis testing (e.g., t-test, ANOVA test, population proportion test), confidence intervals, correlation analysis and regression analysis to make inferences and draw conclusions about data.
- Developed statistical models (regression analysis, panel data modeling) for regional development projects to contribute to data-driven decisions.

### Senior Process Development Analyst an FMCG Company

Jun 2019 — Jan 2022

- Interacted with internal customers to understand business needs and translate into requirements and project scope.
- Assessed the impact of current business processes on users and stakeholders and evaluated potential areas for improvement.
- Maintained strong working knowledge of ERP (SAP), CRM and business intelligence tools and operational features.

### Internal Auditor at a Telecommunication Company

Nov 2015 — Jun 2019

- Performed strategic planning, execution and finalization of audits using data analytics and critical thinking skills.
- Investigated discrepancies discovered during the auditing process.
- Recommended new methods to improve internal controls and operating efficiency.

## RESEARCH INTERESTS

---

- Healthcare analytics
- Data-driven decision making
- Statistical learning and Machine learning
- Reinforcement learning and stochastic optimization
- Time series forecasting
- Probabilistic forecasting

## SKILLS & EXPERTISE

---

**Expertise:** Mathematical and Statistical Modeling, Machine Learning, Time Series Analysis and Forecasting, Stochastic Optimization and Reinforcement Learning, Statistical and Explanatory Data Analysis, web scraping in Python

**Programming:**  Python  SQL

**Reporting:**  Quarto  Advanced Excel  QlikView  SAS

## PROFESSIONAL DEVELOPMENT

---

- Certification:
  - Data Science: Machine Learning, HarvardX (edX)
- Books (some of my go-to books):
  - Forecasting: Principles and Practice (Hyndman et al, 2021)
  - Introduction to Statistical Learning with Applications in R (Tibshirani et al, 2019)
  - The Elements of Statistical Learning (Tibshirani et al, 2008)
  - Probabilistic Machine Learning: An Introduction (Murphy, 2022)
  - Reinforcement Learning: An Introduction (Barto et al, 2018)
  - Reinforcement Learning and Stochastic Optimization (Powell, 2022)
  - A Student's Guide to Bayesian Statistics (Lambert, 2018)
  - Dive into deep learning (Zhang, 2022)

## SCHOLARSHIPS & GRANTS

---

- Welsh Graduate School for the Social Sciences (WGSSS) Studentship Award

Aug 2024

- NATCOR Bursaries / EURO Funding for Stochastic Modelling *Mar 2025*
- The Best Presentation Award at the 5th Welsh Postgraduate Research Cluster Workshop *Sep 2025*

## REFERENCES

---

Available upon request